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Stable and Efficient Flexible Semi-Transparent Organic Photovoltaic Modules with Indium Tin Oxide Top Electrode

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Abstract

Transparent top conductive electrodes are pivotal for semi-transparent organic photovoltaics (ST-OPVs) as it dictates internal light-field distribution, interfacial charge-collection kinetics, and the ultimate environmental robustness of the device. Indium tin oxide (ITO) is the most attractive top electrode for ST-OPVs due to its excellent balance among conductivity, transparency and stability. However, integrating ITO top electrode via magnetron sputtering leads to catastrophic plasma damage to fragile organic active layer in OPVs. Herein, we report a universal interface architecture by introducing a discontinuous metal nano-islands charge-injection buffer layer, enabling damage-free ITO sputtering. In the architecture of ST-OPVs, this metal-hole transport layer (M-HTL) combination acts not only as a kinetic scavenger to quench high-energy species but also as an interfacial dipole that eliminates the Schottky barrier for hole extraction. This strategy yields a record light utilization efficiency of 5.08% among ST-OPVs using non-Ag semitransparent top electrode and enables the scalable fabrication of flexible sub-modules with an active area exceeding 340 cm2 and a power conversion efficiency of 10.26%. Furthermore, the unencapsulated devices demonstrate one order-of-magnitude enhancement in operational lifetime compared to the Ag-based counterparts, while the encapsulated flexible sub-modules tolerate severe thermal and moisture stress for over 3700 hours with almost no performance change. This approach provides a generalized solution for integrating industrial ITO processing

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Stable and Efficient Flexible Semi-Transparent Organic Photovoltaic Modules with Indium Tin Oxide Top Electrode

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Abstract

Transparent top conductive electrodes are pivotal for semi-transparent organic photovoltaics (ST-OPVs) as it dictates internal light-field distribution, interfacial charge-collection kinetics, and the ultimate environmental robustness of the device. Indium tin oxide (ITO) is the most attractive top electrode for ST-OPVs due to its excellent balance among conductivity, transparency and stability. However, integrating ITO top electrode via magnetron sputtering leads to catastrophic plasma damage to fragile organic active layer in OPVs. Herein, we report a universal interface

architecture by introducing a discontinuous metal nano-islands charge-injection buffer layer, enabling damage-free ITO sputtering. In the architecture of ST-OPVs, this metal-hole transport layer (M-HTL) combination acts not only as a kinetic scavenger to quench high-energy species but also as an interfacial dipole that eliminates the Schottky barrier for hole extraction. This strategy yields a record light utilization efficiency of 5.08% among ST-OPVs using non-Ag semi-transparent top electrode and enables the scalable fabrication of flexible sub-modules with an active area exceeding 340 cm^2 and a power conversion efficiency of 10.26%. Furthermore, the unencapsulated devices demonstrate one order-of-magnitude enhancement in operational lifetime compared to the Ag-based counterparts, while the encapsulated flexible sub-modules tolerate severe thermal and moisture stress for over 3700 hours with almost no performance change. This approach provides a generalized solution for integrating industrial ITO processing with high-performance organic and hybrid optoelectronics, accelerating commercialization of power-generating smart windows.

Main

Transparent conductive electrodes are essential for next-generation optoelectronics, underpinning applications ranging from building-integrated photovoltaics (BIPV) and tandem solar cells to transparent displays and wearable electronics.[1-9] The performance of these electrodes fundamentally dictates the light-harvesting limits and operational stability of the entire device stack.[10-12]

Semi-transparent organic photovoltaics (ST-OPVs) exemplify the critical trade-offs inherent in transparent top electrode selection.[13,14] These devices function as power-generating windows by harvesting near-infrared (NIR) photons while transmitting visible light, thereby requiring a delicate balance between power conversion efficiency (PCE) and average visible transmittance (AVT).[15,16] Historically, the field has relied on ultrathin silver (Ag) films (8–15 nm) as transparent

top electrodes to achieve a high PCE, or silver nanowires (AgNW) for high AVT.[17-30] However, percolation-limited island growth imposes a transparency ceiling for ultrathin Ag top electrodes, while AgNW electrodes are constrained by high surface roughness. Furthermore, these Ag-based electrodes suffer from fundamental limitations: thermodynamic instability drives oxidative degradation, and large-area non-uniformity induces severe performance losses upon module scaling.[31-36]

These constraints have motivated a decades-long search for alternative electrode architectures such as transparent conductive oxides (**Fig. 1a**).[37] Indium tin oxide (ITO), the industry gold standard, unparallelly combines the advantages of high visible transmittance (**Fig. S1**), low sheet resistance, and exceptional environmental stability (**Fig. S2**).[11,37] However, the integration of ITO as a top electrode in solution-processed ST-OPV presents a formidable challenge: conventional magnetron sputtering deposition—required for high-quality ITO films—inevitably causes catastrophic damage to the underlying fragile organic semiconductors through high-energy particle bombardment and plasma-induced defect formation. Additionally, the high-temperature annealing (> 300 ℃) typically required to reduce ITO sheet resistance is also detrimental to organic active layers.[38] Lunt and Bulovic first demonstrated ITO as a top electrode in ST-OPVs in 2011.[39] By depositing ITO via low-temperature sputtering onto bathocuproine-capped organic layers, they validated the conceptual feasibility of this strategy, achieving a PCE of 1.7% and an AVT of ~55%. Nevertheless, plasma damage incurred during ITO sputtering degrades the fill factor (FF) and open-circuit voltage (V_{OC}) of the resulting ST-OPV. Consequently, reports on ITO top electrode-based ST-OPVs have remained scarce, and their PCE still lag behind those of their Ag-based counterparts.[40-42]

The key scientific challenge lies in elucidating and mitigating the plasma-induced damage mechanism during magnetron sputtering. Herein, we present a universal interface architecture that resolves the long-standing incompatibility between high-performance organic photovoltaics and industrial ITO sputtering processes. Our strategy employs a discontinuous metal nano-islands

charge-injection buffer layer deposited atop the molybdenum oxide (MoO_x) hole-transporting layer (HTL), forming a composite metal-HTL (M-HTL) to enable damage-free, low-temperature ITO sputtering (**Fig. 1b**). This M-HTL architecture fulfills three critical functions simultaneously: (1) acting as a kinetic scavenger that quenches high-energy plasma species before they reach the organic active layers;[43-45] (2) generating an interfacial dipole that reduces the HTL work function,[46] thereby eliminating the Schottky barrier to hole extraction and establishing Ohmic contact with the ITO top electrode; and (3) maintaining optical discontinuity to suppress parasitic absorption and preserve high visible transmittance.[47,48]

The optimized ST-OPV devices fabricated from this strategy achieve a record light utilization efficiency (LUE) of 5.08% among ST-OPVs based on non-Ag electrode, which is comparable to the best-performing ultrathin-Ag devices. Most importantly, the devices simultaneously demonstrate one order-of-magnitude enhancement in operational stability compared to Ag-based counterparts. Furthermore, we successfully translate this architecture to scalable fabrication, demonstrating rigid and flexible mini-/sub-modules with active areas exceeding 340 cm^2 and PCE over 10% (**Fig. 1c**). These flexible sub-modules tolerate combined moisture and thermal stress for over 3700 hours with almost no performance change. The flexible ST-OPV sub-modules are integrated into a self-powered installation (**Fig. 1d**). Upon exposure to ambient light, the modules harvest photon flux to drive a mechanical assembly of rotating figurines. Such living installation reconciles high transparency with durable power output, which provides a practical approach for next generation energy-harvesting windows.

Beyond ST-OPVs, this M-HTL strategy provides a generalized solution for integrating ITO top electrodes or intermediate recombination layers in diverse organic and hybrid optoelectronic systems—such as perovskite solar cells, organic light-emitting diodes, and tandem architectures. Our findings establish a paradigm for reconciling high-performance solution-processed semiconductors with industrial-scale transparent electrode fabrication, accelerating the commercialization of power-generating smart windows and transparent photovoltaics.

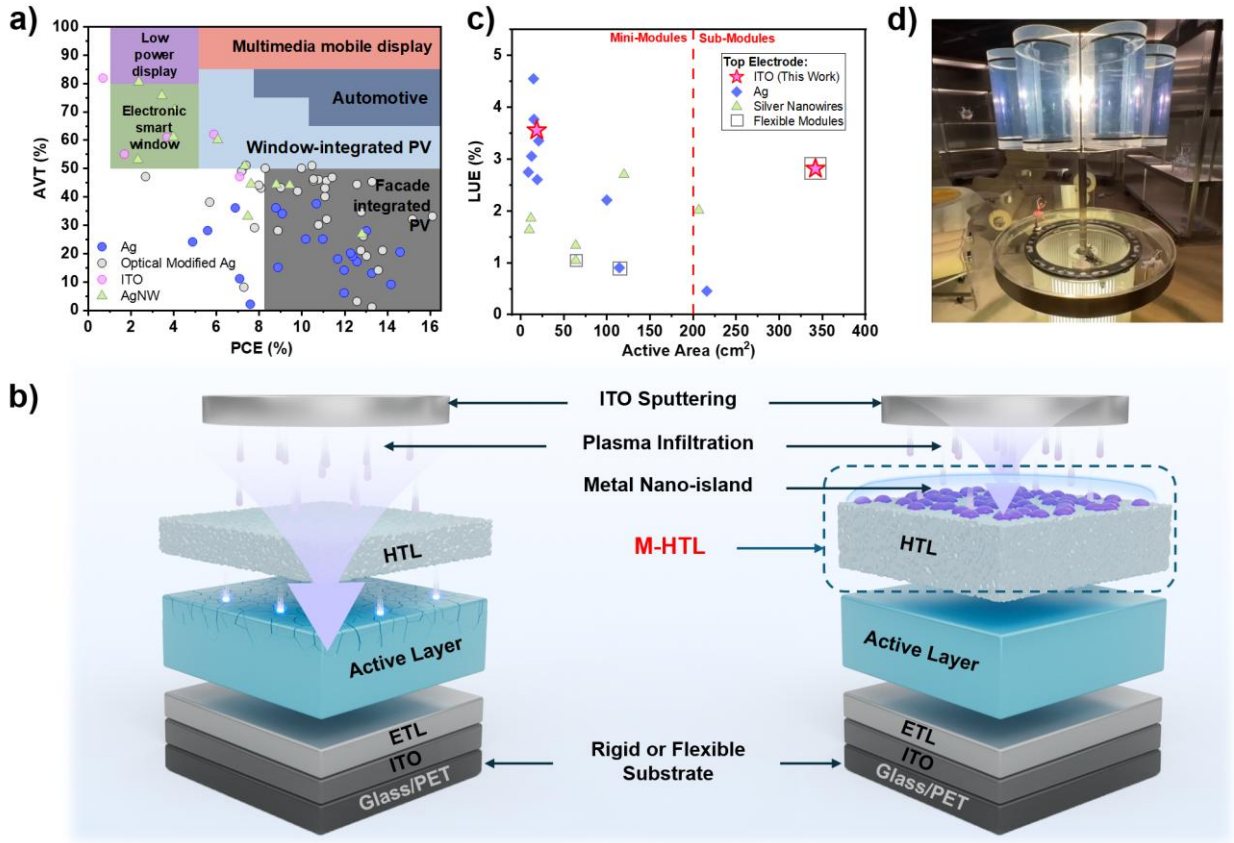


Fig. 1 | ITO-based top transparent electrode for window integrated ST-OPVs. a, Statistics on the AVT and PCE of representative ST-OPVs with different transparent top electrodes, and the possible application scenario requirement according to Lunt, et.al.[1] **b**, Illustration of the M-HTL protecting function during ITO top transparent electrode deposition process. **c**, Statistics of reported LUE of ST-OPV mini- and sub-modules (references are listed in **Table S1**). **d**, Autonomous installation leveraging structural flexibility and transparency. Photograph of a self-powered artware assembly where semi-transparent, flower-shaped flexible modules drive figurine rotation. Short videos are provided in Supporting Information.

Design of the Multi-functional Interface

We fabricated ST-OPVs using the high-performance donor:acceptor blend PM6:L8-BO (chemical structures, **Fig. S3**) with inverted structure of ITO/ZnOnp/PM6:L8-

BO/MoO_x/electrode (**Fig. S4**), wherein ZnO nanoparticles (np) and MoO_x serve as electron-transporting layer (ETL) and HTL, respectively. The MoO_x interlayer, deposited by high-vacuum thermal evaporation, forms an amorphous dense barrier that blocks high energy plasma during ITO growth, while its deep work function (WF) ensures efficient hole extraction from active layer.[49,50] We first established a baseline using thermally evaporated Ag (12 nm) or sputtered ITO (120 nm) as top electrodes on MoO_x with 10 nm thickness commonly used in inverted OPVs.[51] As shown in **Table 1**, ST-OPV device based on the sputtered ITO electrode exhibits a much higher AVT of 43.5% while a much lower PCE of 8.52% than those (17.0% and 14.42%) based on Ag. The higher AVT of ITO-based device is caused by the higher transparency of ITO in the visible region (400–700 nm) (see **Fig. S1**). Directly sputtering ITO onto 10 nm MoO_x layer causes a rapid drop in FF and V_{OC}. Increasing the thickness of MoO_x to 25 nm, the PCE of ST-OPV device increases to 9.87% from 8.52%, while AVT still maintains at 39.3%.

Table 1 | Summarized photovoltaic parameters of ST-OPVs with different HTL and top electrode configurations under AM 1.5G illumination, 100 mW cm^{-2}.

HTL	Top Electrode	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)	AVT (%)	LUE (%)
10 nm MoO_x	12 nm Ag	0.860 (0.856±0.004)	21.5 (20.8±1.2)	78.1 (77.3±1.1)	14.42 (13.73±0.78)	17.0	2.45
10 nm MoO_x	12 nm Ag /40 nm TeO_2	0.854 (0.853±0.002)	20.1 (19.0±1.0)	79.2 (77.8±0.9)	13.61 (12.64±0.75)	26.8	3.65
10 nm MoO_x	120 nm ITO	0.819 (0.835±0.026)	14.2 (13.7±0.5)	73.5 (71.8±3.8)	8.51 (8.20±0.33)	43.5	3.70
25 nm MoO_x	120 nm ITO	0.840 (0.837±0.003)	15.3 (14.7±0.4)	77.0 (77.5±1.1)	9.87 (9.52±0.20)	39.3	3.88
M-HTL (25 nm MoO_x /1 nm Al)	120 nm ITO	0.859 (0.853±0.009)	17.6 (17.5±0.3)	78.7 (77.8±1.1)	11.92 (11.59±0.16)	39.0	4.65

M-HTL (25 nm MoO_x /1 nm Al)	120 nm ITO (with ARC)	0.860 (0.846±0.007)	18.7 (18.5±0.3)	77.8 (77.1±1.0)	12.52 (12.04±0.24)	40.6	5.08

*The average values with standard deviation were calculated from 10 devices.

Optimization of the HTL reveals that increasing the MoO_x thickness provides marginal physical barrier against sputtering, however, thicker films introduce parasitic absorption and series resistance, reducing short-circuit current density (J_{SC}) and FF of OPV devices (**Fig. S5** and **Table S2**). An optimized thickness of 25–30 nm maximizes its blockage effect during ITO deposition without incurring significant optical penalty, yielding an optimal balance between active layer protection and device transparency. To further augment the device performance, an ultrathin metal was inserted as the buffer layer at the MoO_x/ITO top electrode interface. This ultrathin film introduces an interfacial dipole and serves a dual purpose: (1) to establish an intimate, low-resistance contact that mitigates plasma-induced interface defects and (2) to sculpt a graded WF landscape that accelerates Ohmic hole-extraction.[46] Systematic screening of different elemental metals (Ag, Al, Au, Bi) (**Fig. 2a**, **Fig. S6**) identifies Al as the optimal candidate: while Ag and Au upshift the MoO_x WF, Al and Bi downshift it to align with ITO WF (4.40 eV), thereby enhancing efficient hole extraction. Despite Ag shows a high bulk conductivity, it suffers from severe parasitic absorption, whereas Bi's high resistivity—50 times that of Al—stalls carrier transport. In summary, Al provides the highest LUE enhancement (see **Table S3**) and was selected as the adlayer to improve ST-OPV device performance.

The thickness of the Al adlayer dictates the critical balance between interfacial energetics and visible transmittance **(Fig. 2b**, **Fig. S7** and **Table S4**). As Al thickness increases from 1 to 10 nm, a continues metal film grows gradually from discrete small clusters (confirmed by the morphology analysis below). The LUE maximum is reached at 1 nm Al, beyond which the gain in photocurrent is offset by parasitic absorption and a sharp drop in AVT. While PCE remains remarkably stable between 1 and 10 nm Al, this plateau reflects a trade-off between electric beneficial and energy level beneficial factors, which can be observed by the diminishing series

resistance (R_s) and the shifting WF landscape (**Fig. 2c**). Ultraviolet photoelectron spectroscopy (UPS) analysis quantifies this evolution (**Fig. S8** and **Table S5**): as Al thickness increases to 10 nm, the MoO_x surface WF drops from 4.68 eV to 3.79 eV. Crucially, 1 nm Al optimizes this alignment at 4.47 eV, corresponding to an energy level shift of 0.21 eV. This WF shift is a direct spectroscopic signature of metal-oxide interfacial dipole formation (**Fig. 2d**), while charge transfer occurs from low WF metal (Al) surfaces to high WF bulk oxide (MoO_x).[46] The 1 nm Al covered surface WF near-perfectly matches the ITO Fermi level (4.40 eV)—whereas thicknesses beyond 2 nm overshoot this target, creating a new barrier to hole extraction.

Surface analysis confirms that the 1 nm ultrathin Al condenses into discrete nano-islands rather than a continuous film (**Fig. 2e**), acting primarily as a moderator of interfacial energetics rather than a lateral conductor. The scale of discontinues Al particle size is around 35 nm, but with ultrathin thickness less than 2 nm, forming quasi-2D nano-islands which are the characteristic morphology of the Volmer–Weber growth mode commonly observed in metal thin films deposited on oxide substrates.[52,53] Atomic force microscopy (AFM) reveals that this nano-islands deposition planarizes the MoO_x surface, reducing the root-mean-square roughness from 1.501 nm to 1.426 nm to foster a more intimate HTL/electrode contact (**Fig. S9**). The resulting stepwise energy cascade (PM6 HOMO -5.20 eV → MoO_x bottom -4.68 eV → MoO_x/Al surface -4.47 eV → ITO -4.40 eV) provides a smooth, unimpeded path for hole transport (**Fig. 2f**). By reconciling high transparency with improved energy level alignment, the M-HTL-stabilized architecture enhances the LUE of ST-OPVs with ITO top electrodes.

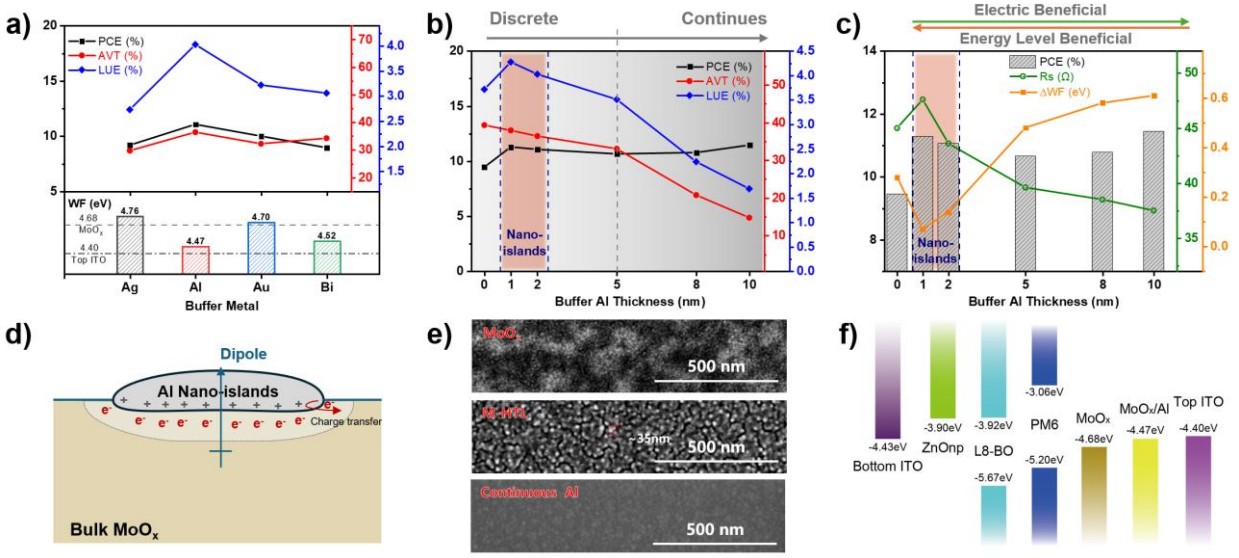


Fig. 2 | Properties of M-HTL layers. a, Screening of metal elements for the buffer layer of ITO-based ST-OPVs. Device structure: ITO/ZnOnp/Active Layer/MoO_x (30 nm)/M (2 nm)/ITO (120 nm). **b**, Performance of ITO-based ST-OPVs with various buffer Al thicknesses. **c**, Relationship of PCE, series resistance (Rs) and work function difference (ΔWF, calculated by $|WF-WF_{topITO}|$) to buffer Al thickness. Device structure for **b** and **c**: ITO/ZnOnp/Active Layer/MoO_x (30 nm)/Al (0-10 nm)/ITO (120 nm). **d**, Illustration of discontinues Al nano-islands forming charge-injection dipole on the surface of MoO_x. **e**, SEM images of the surfaces of bare MoO_x, M-HTL (1 nm Al) and 10 nm continues Al buffer layers. **f**, The energy level diagram of each layer from ITO-based ST-OPV. Energy levels of PM6 and L8-BO are taken from UPS and IEPS results reported in reference[54].

Unraveling the Plasma Protection Mechanism

Fig. 3a schematically depicts the magnetron sputtering process and the potential plasma-induced damage pathways. Ar^+ ions are accelerated toward ITO target, ejecting plasma state high-energy species that impinge on the substrate with kinetic energies reaching hundreds of electron volts.[55] To isolate the mechanism of sputtering damage, we performed ex-situ analysis on organic surfaces after removing the top electrode stacks. Four test samples (**Fig. 3b**) are prepared: "Blank"

denotes a pristine active layer; “Optimized” and “Control 1” correspond to active layers protected by M-HTL (25 nm MoO_x/1 nm Al) and HTL (10 nm MoO_x) during ITO growth, respectively; “Control 2” is a complete device (120 nm ITO on 25 nm MoO_x/1nm Al). The cross-sectional focused ion beam-scanning electron microscope (FIB-SEM) images (**Fig. S10**) proves the complete removal of top ITO electrode along with the M-HTL and MoO_x interlayers in “Optimized” and “Control 1” samples. From their AFM images of the active layers following the selective removal of the top electrode stacks (**Fig. S11**), no obvious morphology changes and topographic roughness are observed.

We first investigated whether high-energy ions physically penetrate into MoO_x oxide lattice. Conventional energy-dispersive X-ray spectroscopy (EDS) employs 7–20 kV high energy electrons that rapidly decompose the organic layer. Instead, high-precision windowless SEM-EDS was performed at 2 kV to minimize electron beam damage and to resolve the Indium M line (0.36 keV) from the Nitrogen K signal (0.38 keV). From **Fig. 3c**, while the "Control 2" device shows clear Indium signatures from the top ITO electrode, the "Blank", "Optimized", and "Control 1" surfaces are spectrally identical, allowing us to rule out Indium- or Tin-ion implantation as the primary damage origin.

UPS analysis instead identifies a distinct electronic signature of damage (**Fig. 3d** and **Table S6**). While the M-HTL protected “Optimized” samples show almost identical signal as “Blank”, the “Control 1” exhibit a downward WF shift and elevated Fermi energy (E_F), coupled with the total disappearance of the shallow gap states characteristic of native PM6:L8-BO (noted as E_1 near E_F). Such difference originates from the change on the interface between active layer and HTL. During sputtering, high-energy electrons carried by ITO plasma penetrate the thin MoO_x layer and accumulate at the active layer/MoO_x interface, inducing unintentional n-type doping of the acceptor material (see discussion below). X-ray photoelectron spectroscopy (XPS) analysis (**Fig. 3e** and **Fig. S12**) corroborates this electronic degradation mechanism at the molecular level. The C1s and S2p peaks of “Control 1" exhibit a systematic shift toward lower binding energy

(0.2~0.3 eV) compared to both "Blank" and "Optimized". The C1s spectrum of the "Control 1" also exhibits an increased intensity ratio between the 284.7 eV component (lower oxidation states such as “C=C” or “C–C”) and the 287.0 eV component (high oxidation states such as “C–O” and “C=O”), indicating a decrease in the average oxidation state of carbon within the conjugated molecular skeleton and confirming electron injection–induced electron accumulation (n-type doping) at the molecular level, consistent with the 0.3 eV downward WF shift observed in UPS.

To spatially corroborate these spectroscopic observations, Kelvin probe force microscopy (KPFM) was employed to map the surface potential distribution (**Fig. S13**). The "Control 1" with 10 nm MoO_x exhibits a markedly elevated average surface potential (230 ± 4.6 mV) relative to the "Blank" reference (171 ± 4.4 mV), consistent with the UPS-detected downward WF shift and confirming plasma-induced n-type doping of the surface. The sample with thicker MoO_x layer (25 nm MoO_x as HTL during ITO growth) displays an intermediate potential (154 ± 4.4 mV) lying between "Blank" and "Optimized," indicating that MoO_x layer only partially suppresses the plasma-induced interfacial charging. However, the "Optimized" with M-HTL shows the lowest average potential (111 ± 5.3 mV), indicating that M-HTL layer not only prevents WF degradation but creates an interfacial dipole that lowers the surface potential. The progressively decreasing surface potential from “Control 1” to “Optimized” establishes a direct correlation between M-HTL protection and surface electronic structure integrity.

The above results reveals that the penetration of hot-electrons during ITO magnetron sputtering introduces additional free charge carriers, thereby increasing the electron concentration at the interface and weakening the built-in electric field of the device, reducing the efficiency of photogenerated charge carrier separation and collection, ultimately manifesting as significant losses in both J_{SC} and V_{OC}. The M-HTL layer does not act as a physical wall against heavy ions but as an electrically dense dissipation barrier. Specifically, the underlying dense MoO_x layer provides the baseline mechanical shielding against atomistic bombardment, while the quasi-2D Al nano-islands serve as macroscopically effective, electrostatically cooperative traps that rapidly

thermalize incoming hot electrons across the interconnected localized dipole fields. By blocking plasma and quenching hot-electron energy before it reaches the active layer, the M-HTL preserves the native energy landscape and enables the high-performance integration of ITO electrodes in soft optoelectronics.

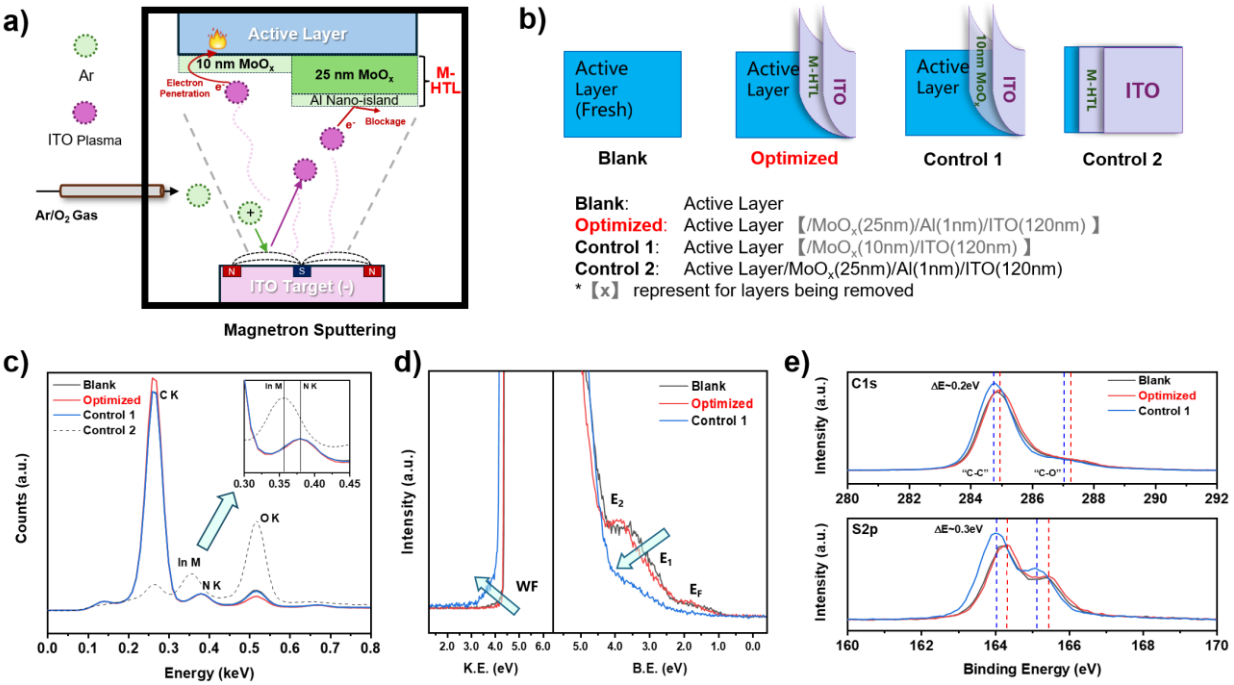


Fig. 3 | M-HTL protecting mechanism. a, Schematic illustration of the high-energy species damage pathways during magnetron sputtering process and the M-HTL protecting mechanism. **b**, Four samples used for ex-situ damage mechanism evaluation. **c**, Windowless SEM-EDS spectra of the surfaces from the mentioned four samples (Accelerate voltage V_{acc}= 2 kV). **d**, UPS and **e**, XPS spectra of the active layer surfaces from blank, optimized and control 1 samples.

Multi-parametric Optical Optimization

To quantify the influence of ITO electrode thickness on device energetics, we performed rigorous Transfer Matrix Method[56-59] simulations based on the device structures shown in **Fig. S4**. The refractive indices (n) and extinction coefficients (k) of the constituent layers are provided in **Fig. S14**.

The calculated light field intensity ($|E|^2$) distributions (**Fig. 4a** and **Fig. S15**) and the corresponding spatial charge generation rate (G_x) (**Fig. 4b** and **Fig. S16**) reveal that the top ITO thickness serves as a critical optical cavity tuner. This structural modulation dictates the transmittance spectra (**Fig. 4c**), where an increasing on top ITO thickness induces a systematic redshift of the visible light transmittance peak from 450 to 600 nm, albeit with reduced peak intensity. Our model predicts a non-monotonic evolution of the AVT, which exhibits a characteristic rise beyond 60 nm to reach a maximum of ~39% at 130 nm (**Fig. 4f**). Excessive thickness, however, triggers destructive interference and parasitic absorption. Such trend is corroborated by the experimental shift in transmittance peaks (**Fig. 4d**). By integrating $|E|^2$, the light percentage distribution of incident light among different optical layers is calculated and shown in **Fig. 4e**. Considering light incident to ST-OPV, photons trapped in active layer could be transferred into photocurrent, visible photons transmitted through the device accounts for AVT, while photons absorbed by the other layers are counted as parasitic absorption.

To simulate and evaluate the influence of ITO electrode on ST-OPV photovoltaic performance, we introduced theoretical maximum achievable $J_{SC,max}$ and theoretical LUE_{max}. $J_{SC,max}$ assumes all absorbed photons can be transferred into photocurrent while LUE_{max} assumes V_{OC} and FF are not influenced by optical parameters so that LUE can be estimated. Both parameters are calculated by **Equation. 1**. The corresponding results are shown in **Fig. 4f**.

$$J_{SC,max} = \int_0^t \int_{300}^{1100} qG(x,\lambda)dxd\lambda$$
$$\mathrm{LUE}_{max} = \mathrm{AVT} \times J_{SC,max} \times V_{OC} \times FF \quad (1)$$

where q is the elementary charge, G (x, λ) is the charge generation rate distribution at position x inside the active layer at a certain wavelength λ, t is the active layer thickness. V_{OC} and FF use 0.86 V and 77% according to the experimental ST-OPV photovoltaic performances shown in **Fig. S17** and **Table S7**. $J_{SC,max}$ of ST-OPVs reaches its peak at the thickness of 90 nm, confirming that the top ITO electrode acts not only as electrode but also an optical modulation layer. Combining with the AVT trend, the LUE_{max} has a maximum value at 120 nm, which can simultaneously

realize the ideal effect of visible transmittance improvement and photocurrent management, optimizing the performance of ST-OPVs. This alignment of the experimental AVT (**Fig. 4f**) with the simulated trajectory confirms that a 120 nm ITO top electrode effectively reconciles broadband conductivity with optimized transparency, providing a robust roadmap for high-performance ST-OPVs.

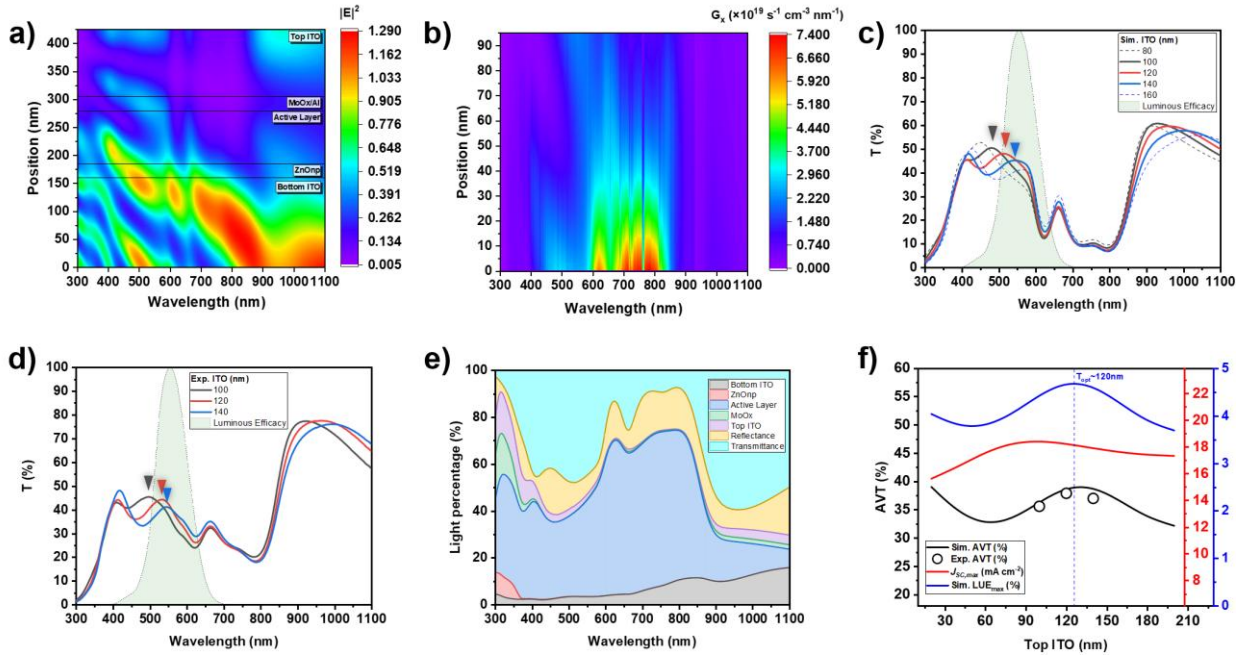


Fig. 4 | Optical simulation. Simulated results on various top ITO thicknesses: **a**, light field intensity distribution across ST-OPV device; **b**, charge generation rate distribution inside active layer; **c**, transmittance spectra. **d**, Experimental transmittance spectra of ST-OPVs with various top ITO electrode thicknesses. **e**, Light percentage distribution among different layers inside ST-OPVs with 120 nm ITO top electrode. **f**, Relationship of simulated AVT, experimental AVT, maximum theoretical current density ($J_{SC,max}$) and theoretical LUE_{max} versus top ITO electrode thicknesses.

Photovoltaic and Optical Performance

The synergistic integration of the M-HTL and sputtered ITO top electrode yields a robust device architecture that simultaneously optimizes carrier extraction and photon flux. Cross-

sectional FIB-SEM images (**Fig. 5a** and **Fig. S18**) confirm the precise stacking of the functional layers, validating the integrity of our vacuum-processed interface. Compared to the commonly used thin 10 nm MoO_x and the thickened 25 nm MoO_x control devices, the M-HTL-based device exhibits a pronounced enhancement in both J_{SC} and V_{OC} (**Fig. 5b**). The coupled enhancement in J_{SC} and V_{OC} reflects a recovered balance between charge generation and recombination. This electrical recovery proves that M-HTL effectively quenches high-energy species during magnetron sputtering, preserving the pristine electronic environment of the underlying organic blend.

To further minimize optical losses and maximize LUE, an anti-reflection coating (ARC) was applied on the light incident surface.[60,61] Reflection spectra (**Fig. S19**) show a broadband suppression of reflectance ($\Delta R \approx 2–4$ %) across the visible range, while the reflectance of the top ITO side remains unchanged. The reduced reflection increases the photon flux available for both absorption and transmission, as evidenced by a modest rise in EQE (**Fig. 5c**) and a concurrent enhancement in transmittance (**Fig. 5d**). Chromaticity parameters are listed in **Fig. S20** and **Table S8**. The optimized device achieves an exceptional sum of EQE and transmittance exceeding 90% across the 500–600 nm spectral window, which precisely aligns with the human photopic sensitivity peak. Furthermore, the devices sustain high aesthetic quality with a color rendering index (CRI) of ~74% and light cyan color coordinates. The total optical balance (EQE + T + R < 100%) across the full spectrum confirms the high precision and internal consistency of our characterization (**Fig. 5e**).[62]

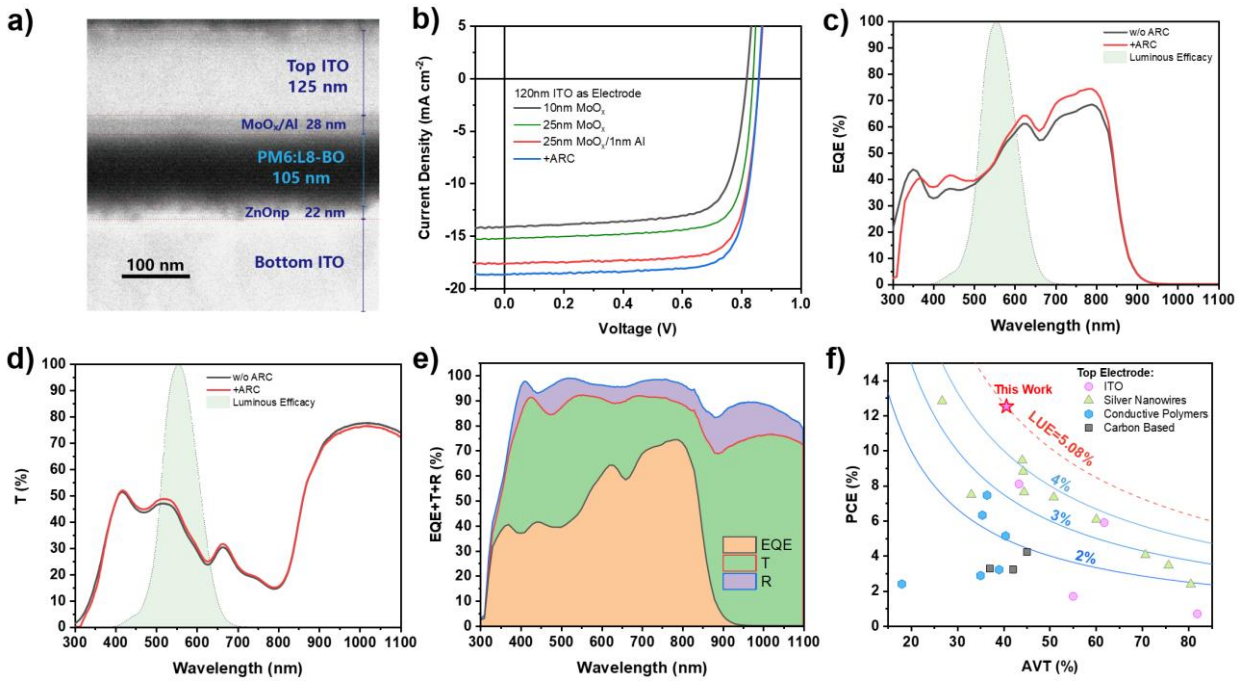


Fig. 5 | Photovoltaic and optical performance of ST-OPVs. a, Cross-sectional FIB-SEM image of ITO-based ST-OPV. **b**, *J-V* curves of ITO-based ST-OPVs under AM 1.5G illumination. **c**, EQE spectra and **d**, transmittance spectra of optimized ITO-based ST-OPVs. **e**, Optical consistency check (EQE+T+R) spectra of ITO-based ST-OPV with ARC. **f**, State-of-the-art LUE performance of this work compared to reported ST-OPVs utilizing non-Ag top electrodes. (References are listed in **Table S9**)

A comprehensive evaluation of the inverted ST-OPV configurations (**Table 1**) underscores the transformative impact of the M-HTL architecture. Control group devices employing 12 nm Ag electrode, which is used commonly in the field,[17,23,24,63-65] suffering from a constrained AVT of 17.0%, limiting their LUE to a mere 2.45% (*J-V* curves and transmittance spectra in **Fig. S21**). Although a 40 nm TeO_2 optical coupling layer boosts the AVT to 26.8%, this additional vacuum processing still fails to meet the stringent transparency requirements for high-end see-through applications.

Transitioning to 120 nm sputtered ITO top electrodes dramatically improves the AVT to 43.5%. However, direct sputtering onto 10 nm MoO_x layer induces severe plasma damage: the

resulting PCE plummets to 8.51%—comparable to the Ag control in LUE (3.70%) but plagued by erratic reproducibility and large standard deviations. The increase of MoO_x thickness itself partially recovers sputtering-induced losses in all three parameters, resulting in a LUE of 3.88%. However, the incorporation of 1 nm Al nano-islands fully restores the photovoltaic performance. With the M-HTL in place, V_{OC} climbs to 0.859 V, J_{SC} surges to 17.6 mA cm^{-2} and PCE reaches 11.92 % while AVT remains at 39.0%, translating into an LUE of 4.65%, suggesting that the sputtering-induced losses is fully recovered. Integrating a glass-side ARC further suppresses reflection, increasing J_{SC} to 18.7 mA cm^{-2} and AVT to 40.6 % without compromising FF (77.8%). Consequently, the champion device realizes a record LUE of 5.08%, which is the highest value reported for ST-OPVs other than Ag electrode (**Fig. 5f**) and already comparable to state-of-the-art Ag-based devices operating at lower transparency.

Operational Stability and Scalability

The transition from Ag to ITO top electrodes fundamentally redefines the operational durability of ST-OPVs. While Ag films are thermodynamically predisposed to spontaneous de-wetting and oxidative degradation under environmental stress,[66,67] ITO provides a chemically inert, ceramic-grade barrier. In rigorous ambient shelf-life tests (25°C, 50% RH), ST-OPVs based on ITO top electrode show extraordinary stability than traditional Ag top electrode (**Fig. 6a**). Unencapsulated Ag-based device suffered near-total failure within 24 hours, whereas our M-HTL protected ITO-based device retained >83% of their initial PCE after 400 hours. Even the ITO-based devices with non-optimized HTL (10 nm MoO_x and 25 nm MoO_x) exhibit PCE retention of 73% and 78% (**Fig. S22, Fig. S23**). Appling optical coupling layer TeO_2 on Ag top electrode slightly improves stability (**Fig. S24**), however, it still falls behind each of the ITO top electrode-based devices.

Continuous maximum power point tracking (MPPT) under continuous 1 sun illumination (ISOS-L-1 protocol) in ambient air reinforces this conclusion: the ITO-based device with M-HTL exhibits a T_{80} lifetime of 89 hours, which is an unprecedented value for unsealed organic solar cells, when compared to the rapidly degrading Ag-based device (**Fig. 6b**). This exceptional durability identifies the sputtered ITO electrode not merely as a robust conductor, but as a self-encapsulating shield that retards the ingress of moisture and oxygen into the underlying organic stack. These results underscore the technological appeal of sputtered ITO top electrodes for long-lifetime, transparent photovoltaic windows amenable to cost-effective large-scale manufacture.

The inherent uniformity of magnetron sputtering allows M-HTL interface to scale seamlessly from millimetric lab cells to industrial-scale modules. Mini-modules (area at 1–200 cm^2 according to National Renewable Energy Laboratory, NREL) on 5.5×5.5 cm^2 glass substrate were first patterned for evaluating interconnection properties (details in **Supplementary Notes**).[68] As shown in **Table 2**, our champion 15-cell mini-module delivers a PCE of 11.20% and an LUE of 3.55%.

Extending this architecture to flexible polyethylene terephthalate (PET) substrates, we successfully fabricated large-area ST-OPV sub-modules (area at 200–800 cm^2 according to NREL) comprising 66 sub-cells at the total area of 341.88 cm^2 (**Fig. 6c**). After flexible encapsulation, the large module delivers 10.26% PCE at a slightly reduced AVT of 27.7% (**Fig. 6d** and **Fig. S25**). These encapsulated flexible ST-OPV modules achieved an LUE of 2.84%, which is to our best knowledge the first flexible sub-module attempts and the highest reported value for ST-OPV modules at this scale (**Fig. 1c**). Under accelerated aging at 50°C and 90% RH ("5090" test), the ST-OPV sub-modules exhibited an unexpected 4.4% performance enhancement at the first 552 h and decay slightly over 3700 h, while control opaque OPV sub-module with 100 nm Ag top electrode retains 80% of its initial PCE (**Fig. 6e** and **Fig. S26)**. The sub-modules probably underwent a thermodynamic self-optimization: the in-situ low-temperature annealing optimizes

the ITO conductivity over time (similar effect as the 300 °C high-temperature short time annealing used in industry).[38,69]

The ITO-based flexible ST-OPV sub-modules are light weighted, while keeping the ability of power output and transmittance of visible scenery. As demonstrated in **Supplementary Video 1**, the bendable module can power a fan while preserving visible transparency. Upon bending, the modules continue to function normally, with the PCE loss following the reduction in light exposure area **(Fig. 6f)**. Notably, the flexible modules bent vertically outperform those bent parallel under more curved conditions over 0.5 cm^{-1} (**Fig. S27**). This directional performance difference is likely due to the distinct effects of vertical and parallel bending on the P1 and P3 interconnection lines, which are critical for sub-cell connectivity (**Fig. S28**). Vertical bending has minimal impact on these lines, whereas parallel bending causes significant broadening and compression, thereby affecting performance. These findings not only confirm the suitability of our flexible ST-OPV sub-modules for integration into curved building surfaces (with bending curvatures mostly $\leq$ 0.2 cm^{-1}) for BIPV deployment, but also highlight the importance of considering the installation direction of flexible modules, a factor that has been previously overlooked.

To showcase the confluence of functional utility and aesthetic expression, we integrated the flexible ST-OPV sub-modules into a self-powered installation (**Fig. 1d** and **Supplementary Video 2)**. These sub-modules combine optical semi-transparency for visual transparency with mechanical bendability, allowing them to be shaped into intricate curved forms such as floral petals. Photon flux is harvested by ST-OPV modules to directly drive the rotation of mechanical figurines. Angular velocity scales in real time with incident light intensity, enabling decorative platforms to function as responsive environmental light sensors. This dynamic system provides a proof-of-concept for next-generation BIPV, demonstrating that energy-harvesting windows and architectural elements can achieve both high optical transparency and sustained power output. By bridging the gap between soft optoelectronics and industrial-scale coating technologies, this work

offers a clear, production-ready pathway for sustainable, transparent power-generating environments.

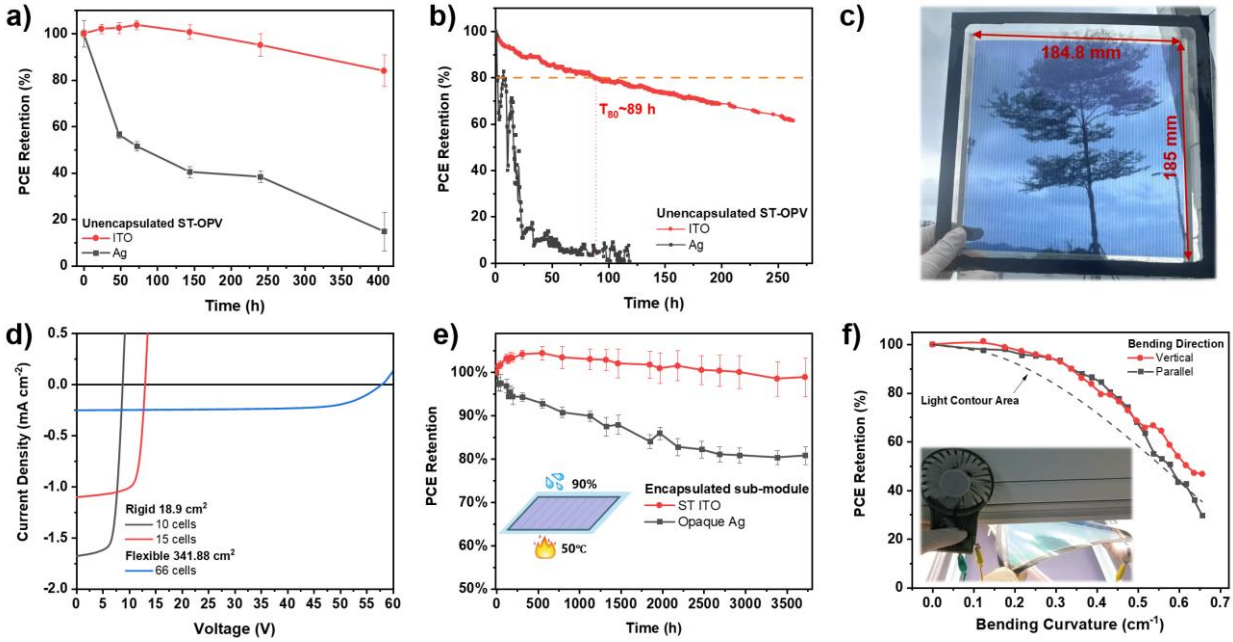


Fig. 6 | Device stability and large-area ST-OPV modules performance. a, Air storage stability and **b**, MPPT operational stability of unencapsulated ST-OPVs based on ITO or Ag top electrodes. **c**, Photograph of observing through the 341.88 cm^2 flexible sub-module. **d**, *J-V* curves of rigid and flexible large area ST-OPV mini- and sub-modules with ITO top electrode. **e**, PCE retention of encapsulated OPV sub-modules with semi-transparent ITO and opaque Ag top electrodes under "5090" condition. **f**, Bending test of encapsulated ITO-based ST-OPV sub-module. Inset: Photograph of a bendable flexible ST-OPV sub-module powering a fan under illumination.

Table 2| Summarized photovoltaic parameters of ST-OPV mini- and sub-modules under AM 1.5G illumination, 100 mW cm^{-2}.

Device	V_{OC} (V)	J_{SC} (mA/cm^2)	FF (%)	PCE (%)	AVT (%)	LUE (%)

Mini-Module 18.9 cm^2	10 cells Rigid	8.80 (8.63±0.07)	1.681 (1.682±0.017)	68.6 (68.2±0.9)	10.15 (9.89±0.21)	31.3	3.18
	15 cells Rigid	13.05 (13.04±0.01)	1.104 (1.080±0.012)	73.1 (72.2±1.2)	11.20 (10.80±0.26)	31.7	3.55
Sub-Module 341.88 cm^2	66 cells Flexible*	58.00 (57.18±0.55)	0.253 (0.257±0.005)	69.9 (68.4±0.8)	10.26 (10.03±0.15)	27.7	2.84

*Flexible module is encapsulated by PET cover.
The average values with standard deviation were calculated from 5 modules.

Discussion and Outlook

Our results demonstrate that magnetron-sputtered ITO top electrodes, enabled by the M-HTL strategy, decisively outperform conventional thin-Ag counterparts in both optical management and operational endurance. The ITO-based architecture achieves superior optical transmittance and enhanced long-term stability, resulting in LUE of 5.08% while maintaining a balanced trade-off between PCE and AVT. To mitigate plasma-induced damage during the ITO sputtering process, an Al-based M-HTL assisted buffer strategy is introduced, and the underlying mechanism is systematically investigated. Notably, the ITO-based ST-OPVs exhibited exceptional operation stability under ambient condition and continuous illumination. Furthermore, we successfully extended this architecture to the fabrication of both rigid and flexible large-area modules. A 341.88 cm^2 flexible sub-module achieved PCE of 10.26% with AVT of 27.7%, which is the best performance ST-OPV in the scale of sub-modules, confirming the scalability and manufacturability of the proposed ITO top electrode approach. These modules exhibit extraordinary resistance to thermal and moisture stress, leveraging the intrinsic chemical stability and self-encapsulating properties of the ceramic ITO electrode. Beyond raw performance, the successful demonstration of transparent windows and kinetic artware proves that ST-OPVs can finally reconcile aesthetic versatility with functional energy harvesting. These achievements

provide a robust, industry-ready roadmap for the rapid integration of organic photovoltaics into the BIPV market.

Data availability statement

The authors declare all data supporting the findings of this study are available within the manuscript and the Supplementary Information.

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Author contributions

H. C., Z. L. and W. Q. conceived the study. S. S. fabricated the solar cells and performed the related measurements. X. Z. fabricated the mini- and sub-modules of semi-transparent solar cells. R. X. performed the optical simulation and analysis. F. L. designed the semi-transparent module demonstration. W. F. fabricated the electron transport layer material. M. S., W. Q., X. Y., A. K.

-Y. J. and H. C. supervised the project. The manuscript was mainly written by S. S., Z. L. and H. C. All the authors contributed to the data analysis and the discussion.

Additional information

Supplementary information is available in the online version of the paper. Reprints and permissions information is available online at www.nature.com/reprints. Correspondence and requests for materials should be addressed to Z. L., W. Q., A. K.-Y. J. and H. C.

Competing financial interests

The authors declare the following competing financial interests:

W. Q., X. Y. and S. S. hold a patent issued to Guangzhou ChasingLight Technology Co., Ltd. regarding the hole transport layer fabricating technology evaluated in this study.

The remaining authors declare no competing financial interests.